

St George Girls High School

Trial Higher School Certificate Examination

2004



Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- All questions may be attempted.
- Begin each question on a new page
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

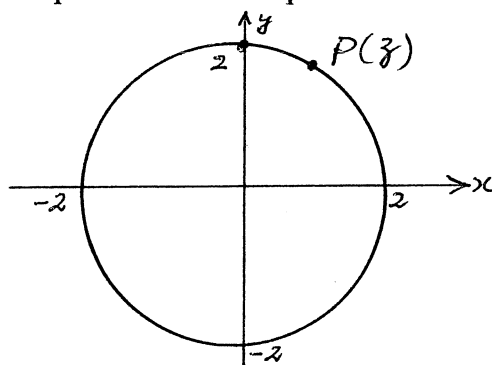
Question 1 – (15 marks) – Start a new page

Marks

a) Evaluate $\int_0^1 \frac{dx}{(x+2)\sqrt{x+2}}$ 2

b) Find $\int \frac{x^2 + 5x - 4}{(x-1)(x^2 + 1)} dx$ 4

c) The point P below represents the complex number z .



(i) Copy the diagram onto your answer booklet.

(ii) By considering both modulus and argument, carefully indicate on your diagram the positions of

a. Q representing $\frac{1}{z}$ 1

b. R representing $\sqrt{z}$ 1

c. S representing $z-1$ 1

d) The locus of all points z in the complex plane which satisfy $\arg\left(\frac{z-3}{z+1}\right) = \frac{\pi}{3}$ forms part of a circle.

(i) sketch this locus. 2

(ii) find the centre and radius of the circle. 2

e) Find $\sqrt{i}$ in the form $a+ib$ where a, b are real numbers. 2

Question 2 – (15 marks) – Start a new page

Marks

a) Find

(i) $\int \frac{dx}{x^2 + 2x + 5}$

1

(ii) $\int \frac{dx}{x\sqrt{x^2 - 1}}$ (using the substitution $x = \sec \theta$)

2

b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{5 + 3 \cos \theta} d\theta$ correct to 3 significant figures.

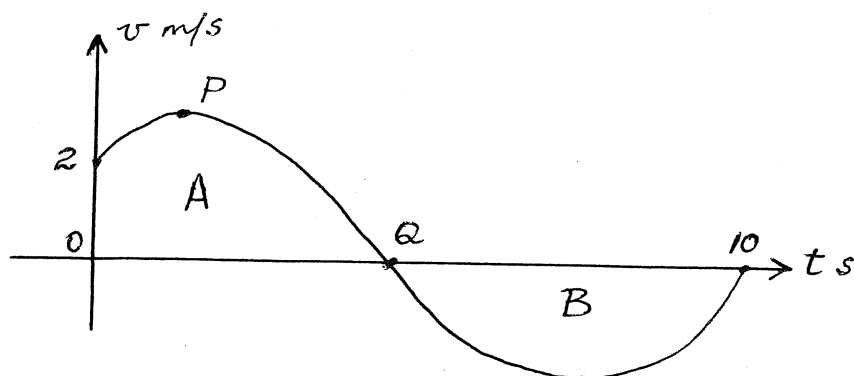
3

c) With the aid of appropriate sketches or otherwise, find the set of values of x for which the limiting sum of the following series exists.

$$1 + \left(\frac{2x-3}{x+1} \right) + \left(\frac{2x-3}{x+1} \right)^2 + \left(\frac{2x-3}{x+1} \right)^3 + \dots$$

4

d) A particle started at the origin with velocity of $2m/s$. Its velocity at any time t ($0 \leq t \leq 10$) is shown below. Area A is 10 and Area B is 15.



(i) Identify two properties of the particle's motion at:

a. the maximum turning point P .

2

b. the point of inflexion Q .

2

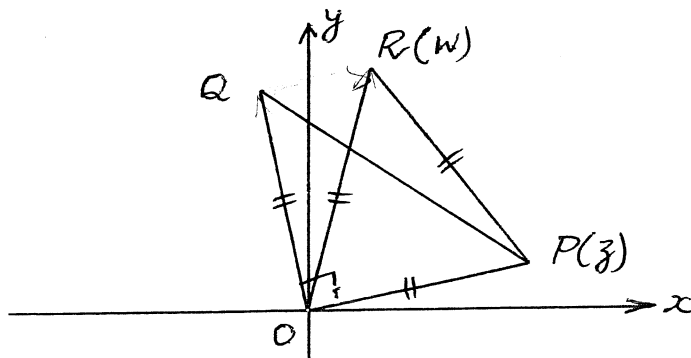
(ii) Where is the particle at $t = 10s$?

1

Question 3 – (15 marks) – Start a new page

Marks

a)



The point P in the complex plane above represents the complex number z . The right angled triangle OPQ is isosceles and the triangle OPR is equilateral.

- (i) Find, in terms of z , the complex number represented by the point Q . 1
- (ii) Find, in terms of z , the complex number which would represent the vector $\overrightarrow{QR}$ 2
- (iii) If R represents the complex number w show that $w^3 + z^3 = 0$. 2

b) Consider the rectangular hyperbola $R : xy = c^2$

- (i) Prove that the equation of the tangent to R at $P\left(ct, \frac{c}{t}\right)$ is $x + t^2y = 2ct$ 2
- (ii) The above tangent crosses the x -axis at M and the y -axis at N . If O is the origin, show that the area of triangle OMN is independent of P . 2

- c) (i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$ 4

- (ii) The line through P parallel to the y -axis meets the asymptote $y = \frac{bx}{a}$ at Q . The tangent at P meets the same asymptote at R . The normal at P meets the x -axis at G . Prove that $\angle RQG$ is a right angle. 2

Question 4 – (15 marks) – Start a new page

Marks

- a) The region bounded by the curves $y = x^2$, $y = (x - 2)^2$ and the x -axis is rotated about the line $x = 2$. Use the method of cylindrical shells to find the volume of the solid. 5
- b) Find the exact value of $\int_0^1 x e^{-x} dx$ 2
- c) Consider the polynomial equation $P(x) = x^4 - 4x^2 + c$
 Find those values of c for which $P(x) = 0$ has
- (i) no real roots. 1
- (ii) 4 different, real roots. 1
- d) Consider the function $f(x) = \frac{x-1}{x}$
- (i) Sketch the graph $y = f(x)$, showing clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes. 2
- (ii) Use the graph $y = f(x)$ to sketch on separate axes the graphs.
- (α) $y = |f(x)|$ 1
- (β) $y = f(|x|)$ 1
- (γ) $y = [f(x)]^2$ 1
- (iii) If $g(x) = \frac{1}{f(x)}$, find $g'(x)$ in terms of $f(x)$ and $f'(x)$, and deduce that the x coordinates of the stationary points of $y = g(x)$ are the x coordinates of the stationary points of $y = f(x)$ for which $f(x)$ is non-zero. 1

Question 5 – (15 marks) – Start a new page

Marks

a) The equation $2x^3 - 3x + 5 = 0$ has roots α, β, γ

(i) Find the polynomial equation with roots of

(α) $\alpha - 1, \beta - 1, \gamma - 1$ 2

(β) $\alpha^3, \beta^3, \gamma^3$ 2

(ii) Evaluate $\frac{1}{\alpha^3} + \frac{1}{\beta^3} + \frac{1}{\gamma^3}$ 2

b) If $z_1 = 1 - \sqrt{3}i$ and $z_2 = \sqrt{3} + i$

(i) Express z_1 and z_2 in modulus-argument form. 1

(ii) Express $\left(\frac{z_2}{z_1}\right)$ in simplest form. 1

(iii) Find the least positive integer n for which z_2^n is real and evaluate z_2^n for this value of n . 2

c) (i) Prove the result:

$$\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$$
 2

(ii) Use this result to show that

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \sin x} dx = 2$$
 3

Question 6 – (15 marks) – Start a new page

Marks

- a) A particle is dropped from rest a height h metres above the ground. At time t seconds its height above the ground, is given by

$$x = h + \frac{gt}{k} + \frac{ge^{-kt}}{k^2} - \frac{g}{k^2}$$

- (i) Show that $\ddot{x} = g - kv$ where the velocity of the particle is v m/s. 2
- (ii) What forces are acting on this particle? Explain carefully. 1
- (iii) If it takes T seconds for the particle to reach half of its terminal velocity, show that $e^{kT} = 2$ 1

- b) Consider the curve C defined by $3x^2 + y^2 - 2xy - 8x - 16 = 0$

- (i) Show that $\frac{dy}{dx} = \frac{3x - y - 4}{x - y}$ 2
- (ii) Find the x coordinates of the points on C where the tangent is parallel to $y = 2x$ 2

- c) If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$, $n \geq 0$

- (i) Show that $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$ for $n \geq 2$ 3

- (ii) Hence, evaluate $\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$ 3

Question 7 – (15 marks) – Start a new page

Marks

a) Find $\int \sec^3 x \, dx$ 3

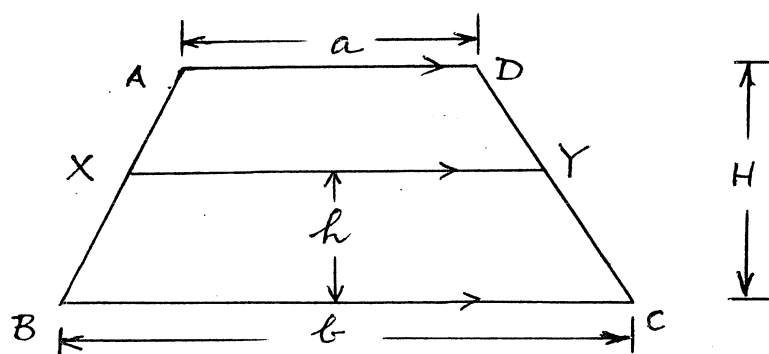
b) Sketch the region R in the Argand diagram consisting of all points z for which:

$$|\arg z| \leq \frac{\pi}{4}, \quad z + \bar{z} \leq 4, \quad |z| \geq 2$$
 3

c) Find the values of the real numbers p and q given that

$$x^3 + 2x^2 - 15x - 36 \equiv (x + p)^2(x + q)$$
 3

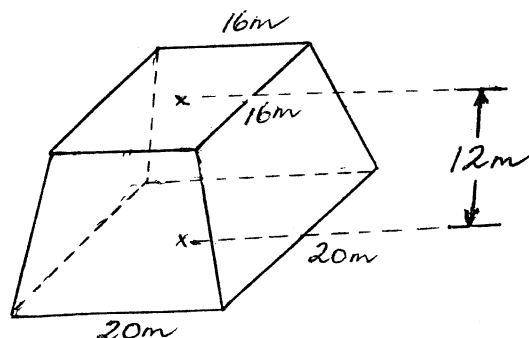
d) (i)



$ABCD$ is an isosceles trapezium of height H with $AB=DC$. The parallel sides AD and BC are of lengths a and b respectively. X and Y are points on AB and DC respectively such that XY is parallel to BC . The perpendicular distance between XY and BC is h . Show that the length of XY is given by:

$$XY = b - \frac{(b-a)}{H} h$$
 3

(ii)



The solid shown has a square base of $20\text{m} \times 20\text{m}$ and a square top of $16\text{m} \times 16\text{m}$.

The top and base lie on two parallel planes. The four sides are isosceles trapezia. The height of the solid is 12m . Find the volume of the solid by taking slices parallel to the base.

Question 8 – (15 marks) – Start a new page

Marks

- a) Find $\int \cos^4 x \, dx$ 3
- b) (i) Find the complex solutions of $z^7 = 1$ 2
and hence factor $z^7 - 1$ over the real numbers. 1
- (ii) Prove that $\cos \frac{3\pi}{7} + \cos \frac{\pi}{7} - \cos \frac{2\pi}{7} = \frac{1}{2}$ 
- c) A particle of mass 3kg moves in a straight line with velocity $v \, m/s$ under a constant force of 5 Newtons and experiences a resistance of $2 + 3v \, N$. The initial velocity of the particle was $v_0 \, m/s$.
- (i) Show that the acceleration, $\ddot{x} \, m/s^2$, is given by $\ddot{x} = 1 - v$ 1
- (ii) Show that $v = 1 - e^{-t} + v_0 e^{-t}$  3
- (iii) Find the terminal velocity. 1
- (iv) When the velocity increases from v_0 to v_1 , show that the distance travelled, $x \, m$, in this time is $x = (v_0 - v_1) + \ln \left(\frac{1 - v_0}{1 - v_1} \right)$ 3

SOLUTIONS.

QUESTION I:

$$\begin{aligned} \text{(a)} \quad \int_0^1 \frac{dx}{(x+2)^{3/2}} &= \int_0^1 (x+2)^{-\frac{3}{2}} dx \\ &= \left[\frac{-2}{(x+2)^{\frac{1}{2}}} \right]_0^1 \\ &= \frac{-2}{\sqrt{3}} + \frac{2}{\sqrt{2}} \\ &= -\frac{2\sqrt{3}}{3} + \sqrt{2} \\ &= \frac{3\sqrt{2} - 2\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{Let } \frac{x^2+5x-4}{(x-1)(x^2+1)} &\equiv \frac{a}{x-1} + \frac{bx+c}{x^2+1} \\ x^2+5x-4 &= a(x^2+1) + (bx+c)(x-1) \end{aligned}$$

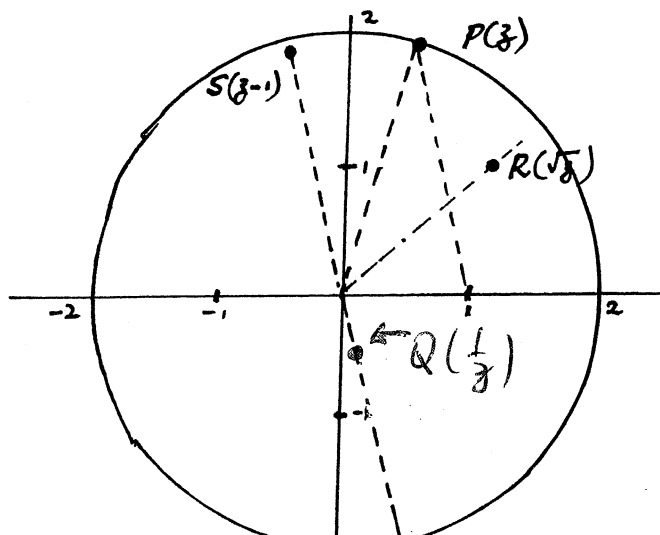
$$\begin{aligned} x=1: \quad 2 &= 2a \\ \therefore a &= 1 \end{aligned}$$

$$\begin{aligned} \text{Co-eff of } x^2: \quad 1 &= a+b \\ &= 1+b \\ \therefore b &= 0 \end{aligned}$$

$$\begin{aligned} \text{Constant:} \quad -4 &= a-c \\ &= 1-c \\ \therefore c &= 5 \end{aligned}$$

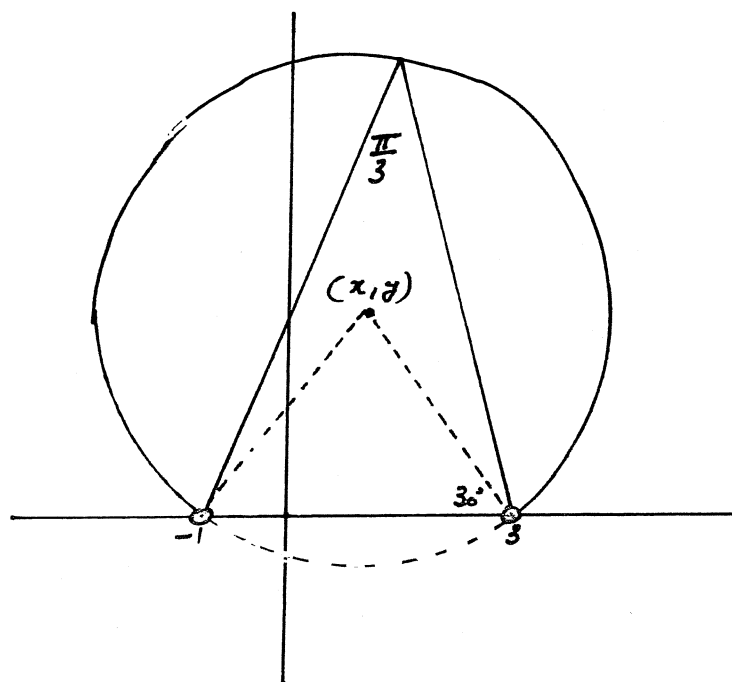
$$\begin{aligned} \therefore \int \frac{x^2+5x-4}{(x-1)(x^2+1)} dx &= \int \left(\frac{1}{x-1} + \frac{5}{x^2+1} \right) dx \\ &= \ln|x-1| + 5 \tan^{-1} x + c \end{aligned}$$

(c)



2.

(d) (i)



$$(ii) \tan 30^\circ = \frac{y}{x}$$

$$y = 2 \tan 30^\circ$$

$$x = 1$$

$\therefore$ Centre is at $(1, \frac{2}{\sqrt{3}})$

$$\cos 30^\circ = \frac{x}{r}$$

$$r = \frac{x}{\cos 30^\circ}$$

$$= \frac{4}{\sqrt{3}}$$

$\therefore$ Radius is $\frac{4}{\sqrt{3}}$

$$(e) \quad i = 1 \operatorname{cis}\left(\frac{\pi}{2} + 2k\pi\right) \quad k \text{ integer}$$

$$\therefore \sqrt{i} = 1 \operatorname{cis}\left(\frac{\pi}{4} + k\pi\right)$$

$$k=0 \Rightarrow \sqrt{i} = \operatorname{cis} \frac{\pi}{4}$$

$$k=1 \Rightarrow \sqrt{i} = \operatorname{cis} \frac{5\pi}{4}$$

$$\text{ie } \sqrt{i} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i, -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$$

QUESTION 2 :

$$(a) \quad (i) \quad \int \frac{dx}{x^2+2x+5} = \int \frac{dx}{(x+1)^2+4} \\ = \frac{1}{2} \tan^{-1}\left(\frac{x+1}{2}\right) + c$$

$$(ii) \quad \int \frac{dx}{x\sqrt{x^2-1}} \quad \begin{array}{l} \text{let } x = \sec \theta \\ dx = \sec \theta \tan \theta d\theta \end{array} \\ = \int \frac{\sec \theta \tan \theta d\theta}{\sec \theta \cdot \tan \theta} \\ = \theta + c \\ = \sec^{-1} x + c$$

$$(b) \quad \int_0^{\frac{\pi}{2}} \frac{d\theta}{5+3\cos \theta} \quad \begin{array}{l} \text{let } t = \tan \frac{\theta}{2} \\ dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta \\ = \frac{t^2+1}{2} d\theta \\ \therefore d\theta = \frac{2dt}{t^2+1} \end{array} \\ = \int_0^1 \frac{\frac{2dt}{1+t^2}}{\frac{5+3(1-t^2)}{1+t^2}} \\ = \int_0^1 \frac{2dt}{5+5t^2+3-3t^2} \\ = \int_0^1 \frac{2}{8+2t^2} dt \\ = \int_0^1 \frac{dt}{t^2+4} \\ = \left[\frac{1}{2} \tan^{-1} \frac{t}{2} \right]_0^1 \\ = \frac{1}{2} \tan^{-1} \frac{1}{2} - 0 \\ = 0.232 \text{ (correct to 3 sig. figs)}$$

$$(c) \quad \text{Series is geometric with } r = \frac{2x-3}{x+1}$$

Limiting sum exists provided $|r| < 1$.

$$\text{ie } \left| \frac{2x-3}{x+1} \right| < 1 \quad x \neq -1$$

$$\text{ie } |2x-3| < |x+1|$$

$$x < -1 \Rightarrow -2x+3 < -x-1$$

$$-1 < x < \frac{2}{3} \Rightarrow -2x + 3 < x + 1$$

$$2 < 3x$$

$$x > \frac{2}{3}$$

$\therefore$ NO SOLUTIONS

$$x > \frac{2}{3} \Rightarrow 2x - 3 < x + 1$$

$$x < 4$$

$\therefore$ Solution is : $\frac{2}{3} < x < 4$.

(d) (i) (a) at P we know that the particle has reached :

- maximum velocity
- zero acceleration
- a point which is less than 10 m to the right of the origin.

(b) at Q the particle has reached :

- maximum deceleration
- zero velocity
- a point which is 10 m to the right of the origin

(ii) at $t = 10$, change in displacement is -5 m

$\therefore$ Particle is 5 m to the left of the origin.

QUESTION 3:

$$(a) \quad (i) \quad P \equiv z$$
$$Q \equiv z \cdot \cos \frac{\pi}{2}$$
$$= iz$$

$$(iii) \quad R \equiv z \cos \frac{\pi}{3}$$

$$\text{ie } w = z \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$\Rightarrow w^3 = z^3 \cos \pi \quad \text{by De-Moivre's theorem}$$
$$= z^3 (-1 + 0i)$$
$$= -z^3$$

$$\therefore w^3 + z^3 = 0$$

$$(ii) \quad (i) \quad Q \equiv iz \quad R \equiv z \cos \frac{\pi}{3}$$

$$\therefore \vec{QR} = z \cos \frac{\pi}{3} - iz$$
$$= z \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i - i \right)$$
$$= z \left[\frac{1}{2} + i \left(\frac{\sqrt{3}}{2} - 1 \right) \right]$$

$$(b) \quad (i) \quad xy = c^2$$

$$\text{diff : } \Rightarrow y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\text{at } P(ct, \frac{c}{t}) \quad \frac{dy}{dx} = \frac{-c/t}{ct}$$
$$= -\frac{1}{t^2}$$

$$\therefore \text{Tangent is } y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$t^2 y - ct = -x + ct$$

$$\text{ie } x + t^2 y = 2ct$$

$$(ii) \quad \text{at } M, y = 0 \quad \therefore M \equiv (2ct, 0)$$

$$\text{at } N, x = 0 \quad \therefore N \equiv (0, \frac{2c}{t})$$

$$\text{Area } \triangle OMN = \frac{1}{2} \cdot 2ct \cdot \frac{c}{t}$$

$= c^2$ which is independent of t
and hence independent of P .

$$(c) \quad (i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{Differentiating} \Rightarrow \frac{2x}{a^2} - \frac{2y}{b^2} \cdot y' = 0$$

$$\frac{2x}{a^2} = \frac{2y}{b^2} \cdot y'$$

$$\therefore y' = \frac{b^2 x}{a^2 y}$$

$$\begin{aligned} \text{at } P(a \sec \theta, b \tan \theta) \quad y' &= \frac{b^2 \cdot a \sec \theta}{a^2 \cdot b \tan \theta} \\ &= \frac{b \sec \theta}{a \tan \theta} \end{aligned}$$

$\therefore$ Equation of normal at P is

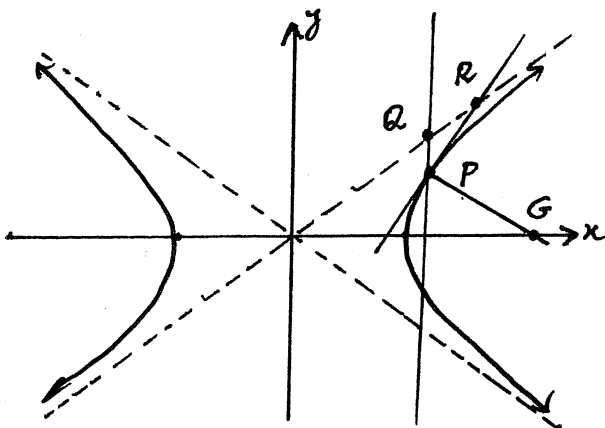
$$y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$\therefore (b \sec \theta) \cdot y - b^2 \sec \theta \tan \theta = (-a \tan \theta) x + a^2 \tan \theta \sec \theta$$

$$\text{ie } (a \tan \theta) \cdot x + (b \sec \theta) y = \sec \theta \tan \theta (a^2 + b^2)$$

$$\sec \theta \tan \theta \Rightarrow \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

(ii)



normal at P crosses x-axis at

$$G \equiv \left(\frac{(a^2 + b^2) \sec \theta}{a}, 0 \right)$$

$$\text{and } Q \equiv (a \sec \theta, b \sec \theta)$$

$$\text{Gradient } QR = \frac{b}{a}$$

$$\begin{aligned} \text{Gradient } QG &= \frac{b \sec \theta - 0}{a \sec \theta - \frac{a^2 + b^2}{a} \sec \theta} \\ &= \frac{ab \sec \theta}{a^2 - a^2 - b^2} \end{aligned}$$

$$= \frac{ab}{-b^2}$$

$$= -\frac{\cancel{b}}{\cancel{b}} - \frac{a}{b}$$

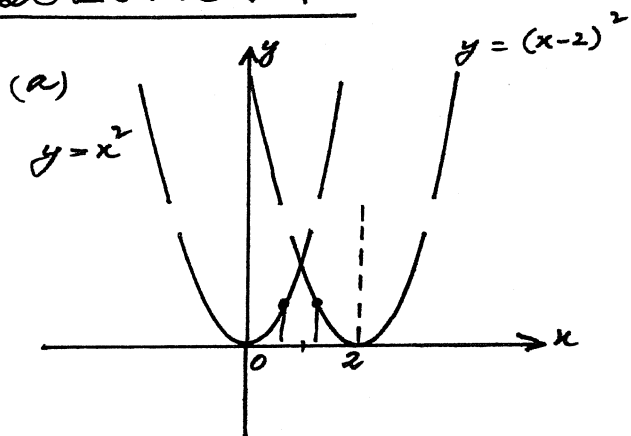
$$\therefore m_{QR} \times m_{QG} = \frac{\cancel{a}}{\cancel{b}} \times \frac{\cancel{b}}{\cancel{a}} \times \frac{b}{a} \times -\frac{a}{b}$$

$$= -1$$

$$\therefore QR \perp QG$$

$$\therefore \hat{RQG} = \frac{\pi}{2}$$

QUESTION 4:



Curves intersect at (1,1)

$$\Delta V_1 = 2\pi(2-x_1) \cdot y_1 \cdot \Delta x \quad \text{where } y_1 = x_1^2 \quad 0 \leq x_1 \leq 1$$

$$\Delta V_2 = 2\pi(2-x_2) \cdot y_2 \cdot \Delta x \quad \text{where } y_2 = (x_2-2)^2 \quad 1 \leq x_2 \leq 2$$

$\therefore$ Volume is given by

$$V = \lim_{\Delta x \rightarrow 0} \left[\sum_{x=0}^1 2\pi(2-x_1) \cdot x_1^2 \Delta x + \sum_{x=1}^2 2\pi(2-x_2)(x_2-2)^2 \Delta x \right]$$

$$= 2\pi \int_0^1 (2x^2 - x^3) dx + 2\pi \int_1^2 (2-x)(x-2)^2 dx$$

$$= 2\pi \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1 - 2\pi \int_1^2 (x-2)^3 dx$$

$$= 2\pi \left[\frac{2}{3} - \frac{1}{4} - 0 \right] - 2\pi \left[\frac{(x-2)^4}{4} \right]_1^2$$

$$= \frac{10\pi}{12} - 2\pi \left[0 - \frac{1}{4} \right]$$

$$= \frac{4\pi}{3}$$

$\therefore$ Volume is $\frac{4\pi}{3}$ units³.

$$(b) \int_0^1 \underbrace{x}_{u} \underbrace{e^{-x}}_{dv} dx$$

$$= x \cdot -e^{-x} \Big|_0^1 - \int_0^1 -e^{-x} dx$$

$$= [-1e^{-1} - 0] - [e^{-x}]_0^1$$

$$= -\frac{1}{e} - \left[\frac{1}{e} - 1 \right]$$

$$= 1 - \frac{2}{e}$$

$$= \frac{e-2}{e}$$

$$(c) \quad P(x) = x^4 - 4x^2 + c$$

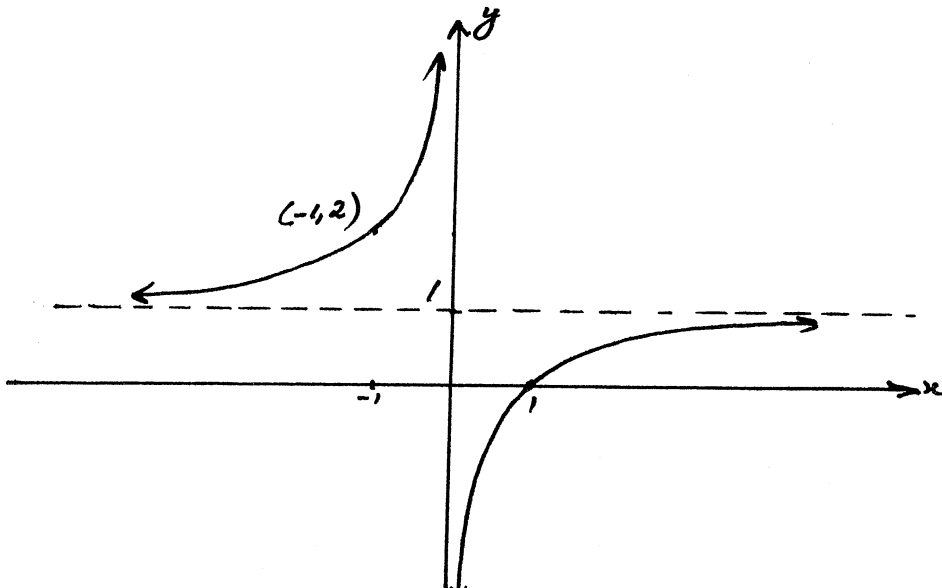
$$= (x^2 - 2)^2 + (c - 4)$$

(i) no real roots $\Rightarrow P(x) > 0$ for all x
and this occurs for $c - 4 > 0$
i.e. $c > 4$

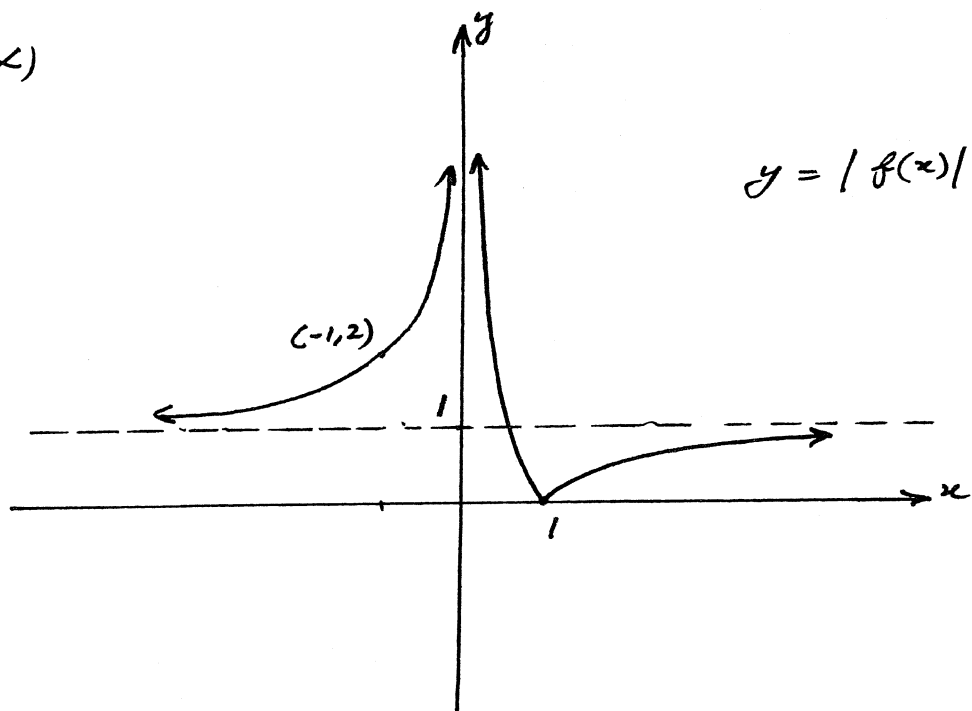
(ii) 4 different real roots $\Rightarrow c - 4 < 0$
 $c < 4$

$$(d) \quad (i) \quad f(x) = \frac{x-1}{x}$$

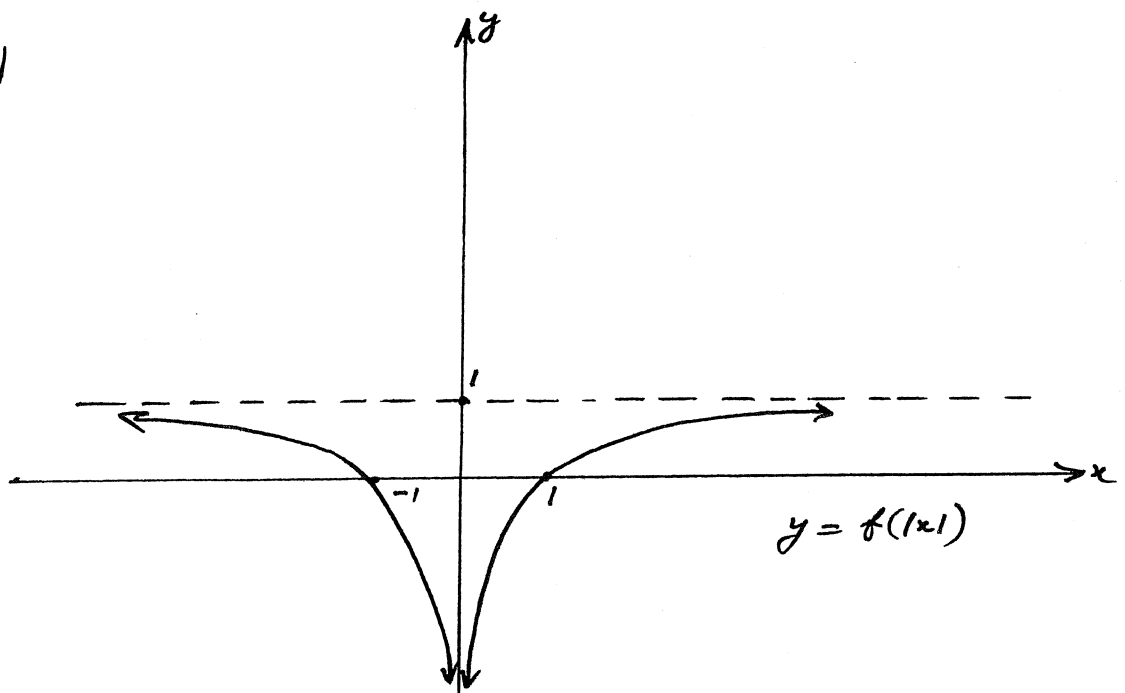
$$= 1 - \frac{1}{x}$$



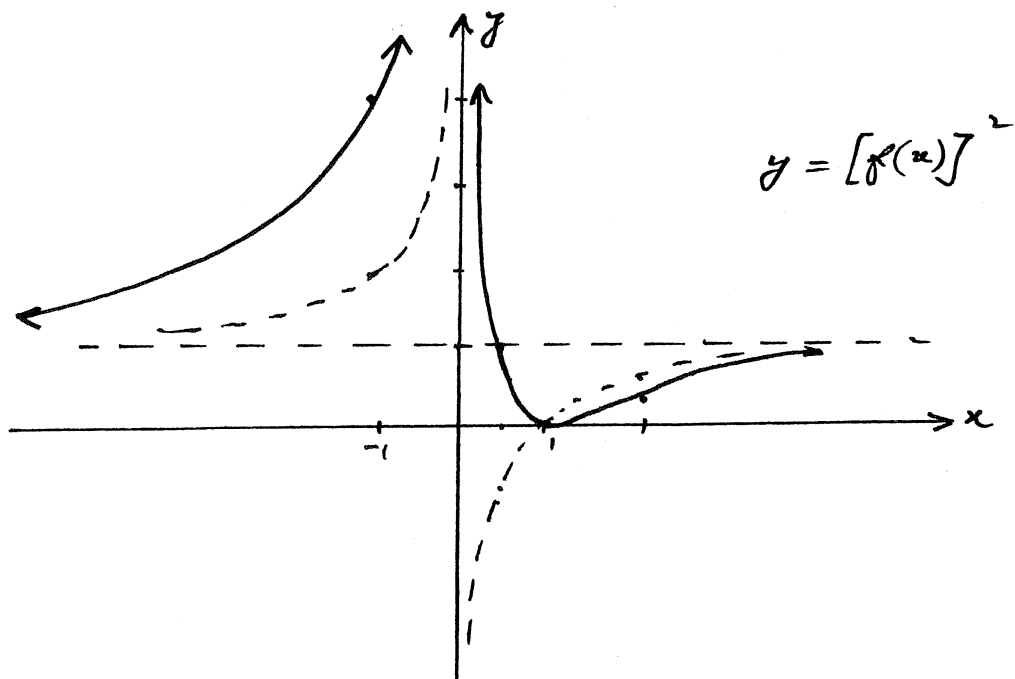
(ii) (α)



(β)



(γ)



$$(iii) \quad g(x) = \frac{1}{f(x)} \\ = [f(x)]^{-1}$$

$$g'(x) = -1 [f(x)]^{-2} f'(x) \\ = - \frac{f'(x)}{[f(x)]^2}$$

Hence $g'(x) = 0$ whenever $f'(x) = 0$
provided $f(x) \neq 0$.

QUESTION 5:

(a) (i) $p(x) = 2x^3 - 3x + 5$

(2) $p(x+1) = 0$

ie $2(x+1)^3 - 3(x+1) + 5 = 0$

$$2x^3 + 6x^2 + 6x + 2 - 3x - 3 + 5 = 0$$

$$2x^3 + 6x^2 + 3x + 4 = 0$$

(3) $p(\sqrt[3]{x}) = 0$

$$\Rightarrow 2x - 3\sqrt[3]{x} + 5 = 0$$

$$2\sqrt[3]{x} = 2x + 5$$

$$27x = 8x^3 + 60x^2 + 150x + 125$$

ie $8x^3 + 60x^2 + 123x + 125 = 0$ has roots $\sqrt[3]{x}$, $\omega\sqrt[3]{x}$, $\omega^2\sqrt[3]{x}$ — (1)

(ii) $\frac{1}{x^3} + \frac{1}{y^3} + \frac{1}{z^3}$
 $= \frac{y^3z^3 + x^3z^3 + x^3y^3}{x^3y^3z^3}$

$$= \frac{123/8}{-125/8} \text{ from (1) above}$$

$$= -\frac{123}{125}$$

(b) (i) $z_1 = 2 \operatorname{cis}(-\frac{\pi}{3})$ $z_2 = 2 \operatorname{cis} \frac{\pi}{6}$

(ii) $\frac{z_2}{z_1} = \frac{2 \operatorname{cis} \frac{\pi}{6}}{2 \operatorname{cis}(-\frac{\pi}{3})}$

$$= \operatorname{cis}(\frac{\pi}{6} + \frac{\pi}{3})$$

$$= \operatorname{cis} \frac{\pi}{2}$$

$$= i$$

$$(iii) \quad z_1 = 2 \operatorname{cis} \frac{\pi}{6}$$

$$z_2^{\sim} = 2^{\sim} \operatorname{cis} \frac{n\pi}{6}$$

$$= 2^{\sim} \left[\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6} \right]$$

For $z_2^{\sim}$ to be real we need $\frac{n\pi}{6} = \pi$

$$\text{ie } n = 6$$

$$\text{Then } z_2^6 = 64 \cdot \cos \pi$$

$$= -64$$

$$(c) \quad (i) \quad \int_{-a}^a f(x) dx = \underbrace{\int_{-a}^0 f(x) dx}_{\text{let } u = -x} + \int_0^a f(x) dx$$

$$= \int_a^0 f(-u) \cdot -du + \int_0^a f(x) dx$$

$$= \int_0^a f(-u) du + \int_0^a f(x) dx$$

$$= \int_0^a f(-x) dx + \int_0^a f(x) dx$$

$$= \int_0^a [f(-x) + f(x)] dx$$

$$(ii) \quad \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1 + \sin x} dx = \int_0^{\frac{\pi}{4}} \left(\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{2}{1 - \sin^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} 2 \sec^2 x dx$$

QUESTION 6:

$$(a) \quad (i) \quad x = h + \frac{gt}{k} + \frac{g}{k^2} e^{-kt} - \frac{g}{k^2} \quad \text{--- (1)}$$

$$\dot{x} = \frac{g}{k} - \frac{g}{k} e^{-kt} = v \quad \text{--- (2)}$$

$$\begin{aligned} \ddot{x} &= g e^{-kt} \\ &= k \left(\frac{g}{k} e^{-kt} \right) \end{aligned}$$

$$= k \left(\frac{g}{k} - v \right) \text{ from (2)}$$

$$\ddot{x} = g - kv$$

(ii) Forces acting are gravity downwards and a resistance force upwards where downwards direction is positive.

(iii) from (2) terminal velocity is $\frac{g}{k}$

$$\therefore (2) \Rightarrow \frac{g}{k} - \frac{g}{k} e^{-kT} = \frac{g}{2k}$$

$$\frac{g}{2k} = \frac{g}{k} e^{-kT}$$

$$\text{i.e. } e^{-kT} = \frac{1}{2}$$

$$\therefore e^{kT} = 2$$

$$(b) \quad C: \quad 3x^2 + y^2 - 2xy - 8x - 16 = 0 \quad \text{--- (1)}$$

(i) Differentiating $\Rightarrow$

$$6x + 2y y' - 2y - 2x y' - 8 = 0$$

$$y'(2y - 2x) = 8 + 2y - 6x$$

$$\therefore y' = \frac{-2(3x - y - 4)}{-2(x - y)}$$

$$= \frac{3x - y - 4}{x - y}$$

$$(ii) \parallel \text{ to } y = 2x \Rightarrow y' = 2$$

$$\therefore \frac{3x - y - 4}{x - y} = 2$$

$$3x - y - 4 = 2x - 2y$$

$$y = 4 - x \text{ sub in } \textcircled{1}$$

$$3x^2 + 16 - 8x + x^2 - 8x + 2x^2 - 8x - 16 = 0$$

$$6x^2 - 24x = 0$$

$$6x(x - 4)$$

$$= 0$$

$$x = 0, 4$$

$$(2) (i) I_n = \int_0^{\frac{\pi}{2}} \underbrace{x^n}_u \underbrace{\sin x}_{dv} dx$$

$$= \underbrace{x^n}_{u} \cdot \underbrace{-\cos x}_{dv} \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\cos x \cdot nx^{n-1} dx$$

$$= n \int_0^{\frac{\pi}{2}} \underbrace{x^{n-1}}_u \underbrace{\cos x}_{dv} dx$$

$$= n \left[x^{n-1} \cdot \sin x \right]_0^{\frac{\pi}{2}} - n \int_0^{\frac{\pi}{2}} \sin x \cdot (n-1) \cdot x^{n-2} dx$$

$$= n \cdot \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2} \quad n \geq 2$$

$$(ii) \int_0^{\frac{\pi}{4}} x^4 \sin x dx = I_4$$

$$= 4 \cdot \left(\frac{\pi}{2} \right)^3 - 4 \cdot 3 \cdot I_2$$

$$= \frac{\pi^3}{2} - 12 \left[2 \cdot \left(\frac{\pi}{2} \right) - 2 \cdot 1 \cdot I_0 \right]$$

$$= \frac{\pi^3}{2} - 12\pi + 24 \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= \frac{\pi^3}{2} - 12\pi + 24 \left[-\cos x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^3}{2} - 12\pi + 24$$

QUESTION 7:

$$(a) \int \sec^3 x \, dx = \int \underbrace{\sec x}_u \cdot \underbrace{\sec^2 x}_{du} \, dx$$

$$= \sec x \cdot \tan x - \int \tan x \cdot \sec x \tan x \, dx$$

$$= \sec x \tan x - \int \sec x \cdot \tan^2 x \, dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx$$

$$= \sec x \tan x - \int \sec^3 x + \int \sec x \, dx$$

$$\therefore 2 \int \sec^3 x \, dx = \sec x \tan x + \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} \, dx$$

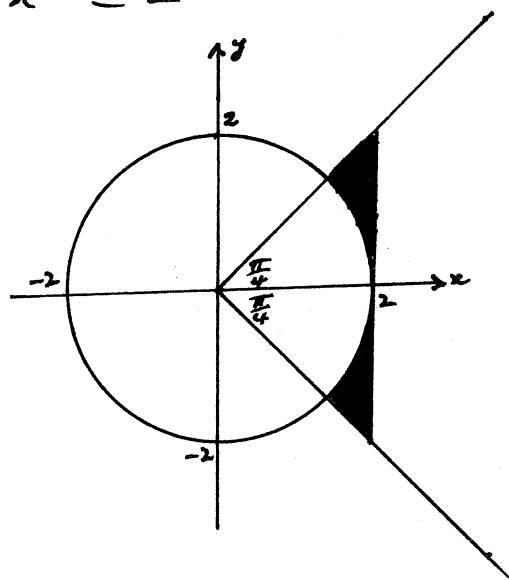
$$\therefore \int \sec^3 x \, dx = \frac{1}{2} \left[\sec x \tan x + \ln |\sec x + \tan x| \right] + C$$

$$(b) \quad z + \bar{z} \leq 4$$

$$\Rightarrow x + iy + x - iy \leq 4$$

$$2x \leq 4$$

$$x \leq 2$$



$$(c) \quad \text{Let } P(x) = x^3 + 2x^2 - 15x - 36 \quad \text{Repeated zero}$$

$$P'(x) = 3x^2 + 4x - 15$$

$$= (x+3)(3x-5)$$

Repeated zero
could be -3 or $\frac{5}{3}$

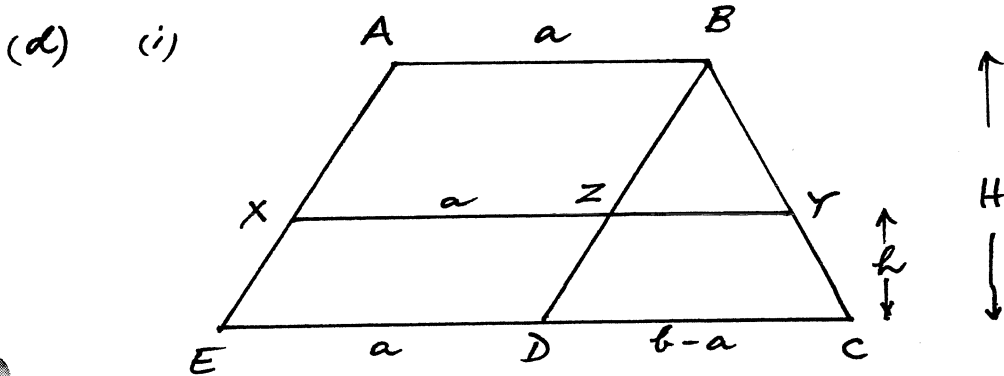
$$P(-3) = -27 + 18 + 45 - 36$$

$$\therefore P(x) = (x+3)^2(x+q)$$

by observation

$$P(x) = (x+3)^2(x-4)$$

$$\therefore p=3, q=-4$$



$\triangle BZY \parallel \triangle BDC$ (equiangular)

$$\therefore \frac{ZY}{b-a} = \frac{H-h}{H}$$

$$ZY = \frac{(b-a)(H-h)}{H}$$

$$\therefore XY = a + \frac{(b-a)(H-h)}{H}$$

$$= a + (b-a) \cdot \frac{H}{H} - \frac{(b-a)h}{H}$$

$$= b - \frac{(b-a)h}{H}$$

(ii) Take a thin slice through the solid parallel to the base and height h m above it.

Volume of slice

$$= \left[20 - \frac{(20-16) \cdot h}{12} \right]^2 \cdot \Delta h \text{ from (i) above}$$

$$= \left(20 - \frac{h}{3} \right)^2 \cdot \Delta h$$

$\therefore$ Volume of solid

$$V = \lim_{\Delta h \rightarrow 0} \sum_{h=0}^{12} \left(20 - \frac{h}{3}\right)^2 \Delta h$$

$$= \int_0^{12} \left(20 - \frac{h}{3}\right)^2 dh$$

$$= \left[\frac{\left(20 - \frac{h}{3}\right)^3}{3 \times -\frac{1}{3}} \right]_0^{12}$$

$$= - \left[16^3 - 20^3 \right]$$

$$= 3904$$

$\therefore$ Volume is 3904 m^3

QUESTION 8:

$$(a) \int \cos^4 x \, dx$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$= \int \left(\frac{\cos 2x + 1}{2} \right)^2 dx$$

$$= \frac{1}{4} \int (\cos^2 2x + 2\cos 2x + 1) dx$$

$$= \frac{1}{4} \int \left(\frac{\cos 4x + 1}{2} + 2\cos 2x + 1 \right) dx$$

$$= \frac{\sin 4x}{32} + \frac{\sin 2x}{4} + \frac{3x}{8} + C$$

$$(b) (i) \quad z^7 = 1 = 1 \operatorname{cis}(0 + 2k\pi) \quad k \text{ integer}$$

$$z = \operatorname{cis}\left(\frac{2k\pi}{7}\right) \text{ by De Moivre's theorem}$$

$$k=0 \quad z = \operatorname{cis} 0 = 1$$

$$k=1 \quad z = \operatorname{cis} \frac{2\pi}{7} = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$$

$$k=2 \quad z = \operatorname{cis} \frac{4\pi}{7} = -\cos \frac{3\pi}{7} + i \sin \frac{3\pi}{7}$$

$$k=3 \quad z = \operatorname{cis} \frac{6\pi}{7} = -\cos \frac{\pi}{7} + i \sin \frac{\pi}{7}$$

$$k=4 \quad z = \operatorname{cis} \frac{8\pi}{7} = -\cos \frac{\pi}{7} - i \sin \frac{\pi}{7}$$

$$k=5 \quad z = \operatorname{cis} \frac{10\pi}{7} = -\cos \frac{3\pi}{7} - i \sin \frac{3\pi}{7}$$

$$k=6 \quad z = \operatorname{cis} \frac{12\pi}{7} = \cos \frac{2\pi}{7} - i \sin \frac{2\pi}{7}$$

$$z^7 - 1 = (z - 1)(z^2 - 2\cos \frac{2\pi}{7}z + 1)(z^2 + 2\cos \frac{3\pi}{7}z + 1)(z^2 + 2\cos \frac{\pi}{7}z + 1)$$

(ii) Sum of roots

$$\Rightarrow 0 = 1 + 2\cos \frac{2\pi}{7} - 2\cos \frac{3\pi}{7} - 2\cos \frac{\pi}{7}$$

$$\therefore \cos \frac{3\pi}{7} + \cos \frac{\pi}{7} - \cos \frac{2\pi}{7} = \frac{1}{2}$$

(c) (i) $m\ddot{x} = F - R$

$$\therefore 3\ddot{x} = 5 - (2 + 3v)$$
$$= 3 - 3v$$

$$\therefore \ddot{x} = 1 - v$$

(ii) $\frac{dv}{dt} = 1 - v$

$$\frac{dt}{dv} = \frac{1}{1-v}$$

$$\therefore t = -\ln(1-v) + c$$

at $t=0$, $v=v_0$

$$\therefore 0 = -\ln(1-v_0) + c$$

$$c = \ln(1-v_0)$$

$$\therefore t = \ln(1-v_0) - \ln(1-v)$$

$$= \ln\left(\frac{1-v_0}{1-v}\right)$$

$$\therefore e^t = \frac{1-v_0}{1-v}$$

$$\frac{1-v}{1-v_0} = e^{-t}$$

$$1-v = e^{-t} - v_0 e^{-t}$$

$$\therefore v = 1 - e^{-t} + v_0 e^{-t}$$

(iii) as $t \rightarrow \infty$ $v \rightarrow 1$

$\therefore$ Terminal velocity is 1 m/s

(iv) $v \frac{dv}{dx} = 1 - v$

$$\frac{dv}{dx} = \frac{1-v}{v}$$

$$\frac{dx}{dv} = \frac{v}{1-v}$$

$$\int_{v_1}^{v_2} \frac{dx}{dv} dv = \int_{v_1}^{v_2} \frac{v}{1-v} dv$$

$$\text{ie } x(v_1) - x(v_0) = \int_{v_0}^{v_1} \frac{v}{1-v} dv$$

$$= \int_{v_0}^{v_1} \left(-1 \cdot \frac{1-v}{1-v} + \frac{1}{1-v} \right) dv$$

$$= \left[-v - \ln(1-v) \right]_{v_0}^{v_1}$$

$$= -v_1 - \ln(1-v_1) - \left[-v_0 - \ln(1-v_0) \right]$$

$$= (v_0 - v_1) + \ln\left(\frac{1-v_0}{1-v_1}\right)$$